REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 2 Lesson 3 | Position Time Graphs

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Graph slope = Δx/Δt; straight-line model x = xi + vt. The graph shows position, not the physical shape of the path.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 37–41.

P01. The slope of a position–time graph represents:

A. Position only B. Elapsed time
C. Mass D. Velocity

Answer

D. Velocity.

P02. A horizontal position–time graph indicates:

A. Rest B. Constant positive speed
C. Constant negative speed D. Increasing acceleration

Answer

A. Rest.

P03. A straight line with negative slope indicates:

A. Positive acceleration B. Constant velocity in the negative direction
C. Increasing position D. Zero velocity

Answer

B. Constant velocity in the negative direction.

P04. Two position–time curves intersect when the objects have:

A. The same velocity necessarily B. The same mass
C. The same position at the same time D. The same distance travelled necessarily

Answer

C. The same position at the same time.

P05. A steeper straight position–time line means:

A. Greater mass B. Longer elapsed time necessarily
C. Smaller velocity magnitude always D. Greater speed magnitude

Answer

D. Greater speed magnitude.

P06. A straight position–time graph joins (0 s, -2 m) and (6 s, 10 m). Calculate its slope.

A. 4 m/s B. 1 m/s
C. 2 m/s D. -2 m/s

Answer

C. 2 m/s. Slope = $\Delta x/\Delta t = (10 - (-2))/6 = 2$ m/s; this is the velocity.

P07. A straight position–time graph is described by x = 4 + 3t, in SI units. Find the position at t = 4 s.

A. 32 m B. 8 m
C. -16 m D. 16 m

Answer

D. 16 m. $x = 4 + 3 \times 4 = 16$ m.

P08. On position–time graphs, A follows xA = 3t and B remains at xB = 24 m. At what time do they meet?

A. 8 s B. 16 s
C. 4 s D. -8 s

Answer

A. 8 s. At intersection, xA = xB. Thus 3t = 24, so t = 8 s.

P09. A position–time graph is horizontal from t = 2 s to t = 10 s. For how long is the object stationary?

A. 16 s B. 8 s
C. 4 s D. -8 s

Answer

B. 8 s. Duration = 10 − 2 = 8 s. A horizontal x–t graph has zero slope and zero velocity.

P10. An object moves from -6 m to 12 m in 8 s. Calculate its average velocity.

A. 4.5 m/s B. 1.125 m/s
C. 2.25 m/s D. -2.25 m/s

Answer

C. 2.25 m/s. $v_{avg} = \Delta x/\Delta t = (12 - (-6))/8 = 2.25$ m/s.

P11. Which quantity belongs on each axis of a position–time graph?

Answer

Time goes on the horizontal axis and position on the vertical axis, with units stated.

P12. How is average velocity found between two graph points?

Answer

Divide the change in position by the corresponding change in time: $v_{avg} = \Delta x/\Delta t$.

P13. What does a curved position–time graph indicate about velocity?

Answer

The slope changes with time, so velocity is not constant.

P14. Why is a position–time graph not a map of the object's path?

Answer

Its axes are position and time, not two spatial coordinates. A sloping graph need not represent motion up a hill.

P15. What graph feature determines initial position at t = 0?

Answer

The position-axis intercept gives the initial position.

P16. A straight position–time graph joins (0 s, -3 m) and (9 s, 15 m). Calculate its slope.

Answer

Slope = $\Delta x/\Delta t = (15 - (-3))/9 = 2$ m/s; this is the velocity.

P17. A straight position–time graph joins (0 s, -4 m) and (12 s, 20 m). Calculate its slope.

Answer

Slope = $\Delta x/\Delta t = (20 - (-4))/12 = 2$ m/s; this is the velocity.

P18. A straight position–time graph is described by x = 6 + 3t, in SI units. Find the position at t = 5 s.

Answer

$x = 6 + 3 \times 5 = 21$ m.

P19. A straight position–time graph is described by x = 8 + 3t, in SI units. Find the position at t = 6 s.

Answer

$x = 8 + 3 \times 6 = 26$ m.

P20. On position–time graphs, A follows xA = 3t and B remains at xB = 36 m. At what time do they meet?

Answer

At intersection, xA = xB. Thus 3t = 36, so t = 12 s.

P21. On position–time graphs, A follows xA = 3t and B remains at xB = 48 m. At what time do they meet?

Answer

At intersection, xA = xB. Thus 3t = 48, so t = 16 s.

P22. A position–time graph is horizontal from t = 3 s to t = 15 s. For how long is the object stationary?

Answer

Duration = 15 − 3 = 12 s. A horizontal x–t graph has zero slope and zero velocity.

P23. A position–time graph is horizontal from t = 4 s to t = 20 s. For how long is the object stationary?

Answer

Duration = 20 − 4 = 16 s. A horizontal x–t graph has zero slope and zero velocity.

P24. An object moves from -9 m to 18 m in 12 s. Calculate its average velocity.

Answer

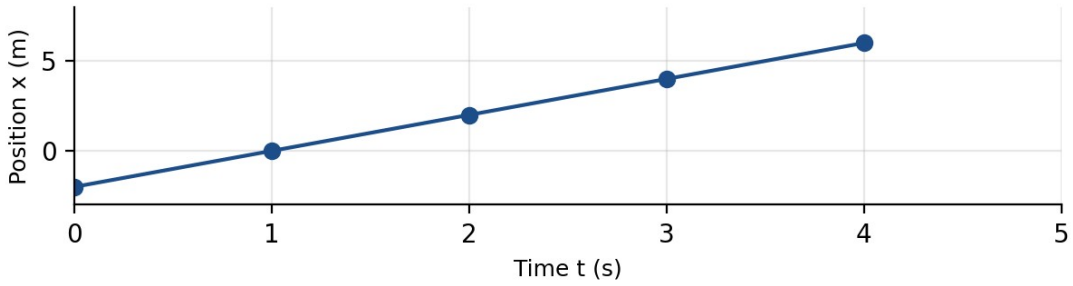
$v_{avg} = \Delta x/\Delta t = (18 - (-9))/12 = 2.25$ m/s.

P25. An object moves from -12 m to 24 m in 16 s. Calculate its average velocity.

Answer

$v_{avg} = \Delta x/\Delta t = (24 - (-12))/16 = 2.25$ m/s.

P26. Plot the data t = 0, 1, 2, 3, 4 s and x = −2, 0, 2, 4, 6 m. Describe the motion.



Answer

A straight line rises from −2 m with slope +2 m/s. The object moves at constant velocity in the positive direction and crosses the origin at 1 s.

P27. Object A follows x = 2t and object B follows x = 12 − t, in SI units. Find when and where their graphs intersect.

Answer

Set 2t = 12 − t: t = 4 s. Then x = 8 m. They share position at that instant, but their velocities differ.

P28. A graph rises, then becomes horizontal, then falls. Describe the motion without assuming numerical speeds.

Answer

The object first moves in the positive direction, then rests, then moves in the negative direction. Slopes determine the speeds.

P29. Can two objects have the same position but different velocities? Explain using graphs.

Answer

Yes. Two lines can intersect while having different slopes. At the intersection the positions agree but the velocities need not.

P30. How should you estimate a graph slope reliably from a best-fit straight line?

Answer

Choose two well-separated points on the fitted line, read their coordinates with units and calculate Δx/Δt. They need not be original measured points.